# ops-knowledge-loop

CONTEXT

# The On-Call Problem

**MEAN TIME TO REMEDIATION**

Mean time to remediation is dominated by lookup time, not fix time

**KNOWLEDGE SILO**

Institutional knowledge lives in engineers' heads, not in systems

**REPEAT WORK**

Repeated incidents get manually triaged every single time

▸ Meridian runs 17 microservices across UK retail infrastructure

ARCHITECTURE

# How It Works

**[01] INCIDENT**
ServiceNow ticket free-text description

→

**[02] RAG SEARCH**
sentence-transformers + ChromaDB top-3 matches

→

**[03] LLM REASONING**
Ollama qwen3:14b → structured JSON response

→

**[04] DECISION GATE**
≥0.70 auto-fire · <0.70 human review

→

**[05] OUTCOME**
Rundeck job fires or engineer gets pre-reasoned brief

▸ Fully local · No data leaves the machine · No API costs · ~10s end-to-end

# What the RAG Searches

## ▸ INCIDENT HISTORY

500+ synthetic incidents across 17 services

Each indexed as a vector embedding in ChromaDB

OOM kills · latency spikes · ETL failures
EDI schema mismatches · POS outages
loyalty bugs · notification duplicates

## ▸ RUNBOOKS

Per-service markdown runbooks

checkout · product-search · payment-processor · …

Auto-generated runbooks from resolved ServiceNow tickets

Chunked and embedded alongside incident history

All embeddings generated with all-MiniLM-L6-v2 · Similarity threshold: 0.60 strong match

TOOLING

# Ops Dashboard

- 📋 Recent incidents and their triage status
- 🎯 RAG match scores per incident
- 🤖 AI recommendation with confidence score
- ⚡ Decision gate outcome (auto-executed vs pending review)
- 🔧 Rundeck job catalogue

`Flask · localhost:5000`

**DASHBOARD PREVIEW**

```
┌──────────────────────────────┐
│ ops-knowledge-loop dashboard │
├──────┬────────────────┬──────┤
│ INC  │ STATUS         │ CONF │
├──────┼────────────────┼──────┤
│ 0043 │ AUTO-EXECUTE   │ 0.95 ✓ │
│ 0044 │ PENDING        │ 0.35 ▪ │
│ 0045 │ AUTO-EXECUTE   │ 0.75 ✓ │
└──────┴────────────────┴──────┘
```

RESULTS

# 10 Queries · Live Run

**5** AUTO-EXECUTED

**5** PENDING REVIEW

gate threshold: 0.70

| INCIDENT | RECOMMENDED JOB | CONFIDENCE | OUTCOME |
|---|---|---|---|
| checkout-service OOM kill | restart-service-with-memory-bump | 0.95 | AUTO-EXECUTE |
| payment-processor 500 errors | rollback-to-previous-version | 0.95 | AUTO-EXECUTE |
| product-search unresponsive | reindex-elasticsearch | 0.75 | AUTO-EXECUTE |
| inventory sync stalled | force-inventory-sync | 0.66 | PENDING REVIEW |
| store POS tills unresponsive | rollback-to-previous-version | 0.75 | AUTO-EXECUTE |
| loyalty service not awarding pts | restart-service-rolling | 0.35 | PENDING REVIEW |
| order fulfilment latency spike | scale-up-replicas | 0.75 | AUTO-EXECUTE |
| notification service dup emails | restart-service-with-memory-bump | 0.35 | PENDING REVIEW |
| supplier EDI orders dropped | force-inventory-sync | 0.65 | PENDING REVIEW |
| recommendation engine degraded | reindex-elasticsearch | 0.45 | PENDING REVIEW |

"The gate is discriminating by design — weak RAG similarity or ambiguous root cause correctly suppresses auto-execution"

LIVE OUTPUT — QUERY 1 / 10

# checkout-service OOM kill · 0.95 → AUTO-EXECUTE

```
$ python query_live.py "checkout service OOM kill, container hitting memory limit and restarting repeatedly under load"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │   0.797 ✓ strong match │ [INC0043096] checkout-service OOM kill   │
│ 2 │ incidents.json │   0.796 ✓ strong match │ [INC0043144] checkout-service OOM kill   │
│ 3 │ incidents.json │   0.795 ✓ strong match │ [INC0043048] checkout-service OOM kill   │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

✓ Strong precedent found — incidents.json (similarity 0.797)

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : restart-service-with-memory-bump
  Confidence       : 0.95
  Decision Gate    : AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : OOM kills due to memory limits directly align with the job purpose.
                     Runbook context explicitly confirms memory threshold breach as trigger.

┌──────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  restart-service-with-memory-bump  (conf 0.95 ≥ 0.70 gate)  |
└──────────────────────────────────────────────────────────────────────────┘
Total runtime: 8.7s
```

LIVE OUTPUT — QUERY 3 / 10

# product-search unresponsive after deploy · 0.75 → AUTO-EXECUTE

```
$ python query_live.py "product search returning empty results, search index unresponsive after deployment"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬──────────────────┬──────────────────────────┬──────────────────────────────────────┐
│ # │ Source           │        Similarity        │ Excerpt                              │
├───┼──────────────────┼──────────────────────────┼──────────────────────────────────────┤
│ 1 │ product-search.md│  0.575 ~ partial match   │ Service Overview: ES-backed search service   │
│ 2 │ incidents.json   │  0.571 ~ partial match   │ [INC0043346] product-search slow query 1352ms│
│ 3 │ product-search.md│  0.570 ~ partial match   │ Diagnostic Steps: kubectl exec cache rate    │
└───┴──────────────────┴──────────────────────────┴──────────────────────────────────────┘

⚠ No close precedent — best match 0.575 in product-search.md. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : reindex-elasticsearch
  Confidence       : 0.75
  Decision Gate    : ABOVE GATE → AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : Unresponsive ES index after deployment maps to reindexing as direct fix.
                     Runbook + incident context both confirm the Elasticsearch-backed architecture.

┌──────────────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  reindex-elasticsearch  (confidence 0.75 ≥ 0.70 gate)      │
└──────────────────────────────────────────────────────────────────────────────────┘
Total runtime: 9.9s
```

LIVE OUTPUT — QUERY 6 / 10

# loyalty service not awarding points · 0.35 → PENDING REVIEW

```
$ python query_live.py "loyalty service not awarding points after purchase, customer accounts not updating"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.642 ~ partial match │ [INC0042145] loyalty-service failing to award pts│
│ 2 │ incidents.json │  0.562 ~ partial match │ [INC0041652] loyalty-service reward API 500      │
│ 3 │ incidents.json │  0.561 ~ partial match │ [INC0041134] loyalty-service negative balances   │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.642. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : restart-service-rolling
  Confidence       : 0.35  ← MODEL SELF-DOWNGRADED
  Decision Gate    : RULE-BASED FALLBACK
  AI Source        : ollama:qwen3:14b
  Reasoning        : Past loyalty-service issues resolved via restarts or config updates.
                     Runbook context references backfill scripts — not a restart path.
                     Semantic mismatch between query and available context. Self-downgraded.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ■  PENDING HUMAN REVIEW  —  restart-service-rolling  (conf 0.35 < 0.70 gate)  |
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 14.6s
```

DESIGN PHILOSOPHY

# The Gate is the Point

**loyalty-service · 0.35**
RAG matched past incidents but runbook referenced a backfill script, not a restart. Model self-downgraded.

**notification-service · 0.35**
Matched incident was push notifications, not email. Semantic mismatch caught.

**supplier-EDI · 0.65**
Schema validation failure; force-inventory-sync is plausible but not certain. Correctly held for review.

```
$ python query_live.py "loyalty service not awarding points after purchase, customer accounts not updating"

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────────┐
│ # │ Source         │       Similarity     │ Excerpt                                      │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────────┤
│ 1 │ incidents.json │  0.642 ~ partial match │ [INC0042145] loyalty-service failing to award pts  │
│   │                │                      │ for click-and-collect — missing order type mapping  │
│ 2 │ incidents.json │  0.562 ~ partial match │ [INC0041652] reward redemption API 500 (DB pool)   │
│ 3 │ incidents.json │  0.561 ~ partial match │ [INC0041134] points returning negatives (bulk event) │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────────┘

⚠ No close precedent — best match 0.642. AI will deduce a remediation.

  Recommended Job  : restart-service-rolling
  Confidence       : 0.35  ← MODEL SELF-DOWNGRADED
  Decision Gate    : RULE-BASED FALLBACK
  Reasoning        : Runbook context references a backfill script — not a restart path.
                     Semantic mismatch between query and available context.

■  PENDING HUMAN REVIEW  —  restart-service-rolling (confidence 0.35 < 0.70 gate)
   On-call engineer sees this recommendation + RAG sources before any action is taken.
Total runtime: 14.6s
```

"A system that knows what it doesn't know is more useful than one that always fires"

TECHNOLOGY

# Stack

**Python 3.11**
Core pipeline

**ChromaDB**
Vector store

**sentence-transformers**
all-MiniLM-L6-v2 embeddings

**Ollama + qwen3:14b**
Local LLM · 9.3GB · Q4_K_M · RTX 4090

**Flask**
Ops dashboard

**Rich**
CLI output formatting

**ServiceNow API**
Mocked ticket source

**Rundeck**
Mocked job execution

▸ Runs entirely on local hardware · Zero external API calls · Zero ongoing cost

ROADMAP

# Where This Goes Next

### Shadow Deployment Pipeline

A parallel shadow environment mirrors all live traffic and incoming incidents without taking any action. When the pipeline's recommendations have earned sufficient confidence, a single approval promotes the shadow's configuration to production and tears down the shadow instance. Zero-downtime, zero-guesswork rollout — the system validates itself before it acts.

### Repository-Aware Context

Continuous indexing of service source repositories gives the LLM access to code-level context — configuration files, dependency changes, recent commits — when an incident's root cause demands investigation beyond historical tickets and runbooks. The knowledge base grows to match the depth of the problem.

### Living Organisational Memory

Automatic ingestion of JIRA tickets, Confluence pages, and equivalent knowledge bases keeps the RAG context current with the team's accumulated decisions, post-mortems, and architectural changes. The system learns continuously from every resolved incident and every documented decision — whether or not it was the one that triaged it.

ROADMAP

# Next on the List

**01 · Feedback Loop**
Engineer overrides become high-signal labelled examples that improve future recommendations automatically.

**02 · Multi-Incident Correlation**
When five services degrade at once, switch to cascading-failure reasoning instead of isolated ticket triage.

**03 · Confidence Calibration**
Use a held-out labelled set and reliability diagram to verify that 0.75 confidence means roughly 75% correct.

**04 · Streaming Output**
Stream reasoning token by token in the dashboard so engineers can interrupt a bad reasoning path early.

▸ Slack / PagerDuty integration brings pending-review recommendations into the engineer's existing workflow.

# Observability

**Log Every Step**
RAG top-K + similarity scores, LLM token counts + latency, every gate confidence and outcome, every Rundeck execution result.

**Dashboards**
Confidence distribution over time, auto-execution rate, false positive rate, and P99 latency per pipeline stage.

**Alerts**
Confidence distribution shift, fallback rate increase, and any Rundeck failure without a subsequent human resolution.

Sudden confidence drops are a production smell: stale knowledge base, model version drift, or a retrieval-quality regression.

PRODUCTION PATH

# From Demo to Production

| 01 | 02 | 03 | 04 |
|---|---|---|---|
| ServiceNow mock→real webhook API | ChromaDB local→managed vector store | Ollama local→hosted or private LLM endpoint | single demo run→queued concurrent triage |

▸ Retrieval at scale improves recall first, then needs mitigation: reindex with better embeddings, filter by service namespace, and combine vector search with BM25.

SUMMARY

# What This Demonstrates

- 01 Production-pattern RAG pipeline with real similarity scoring
- 02 LLM structured output with confidence-gated automation
- 03 Human-in-the-loop safety by design, not by accident
- 04 Extensible to real ServiceNow + Rundeck with config changes only

APPENDIX

# Additional Terminal Outputs

▸ Seven extra incident runs are kept here for deeper Q&A if the manager wants to inspect the wider behaviour set.

Main walkthrough kept **Q1**, **Q3**, and **Q6**; appendix keeps Q2, Q4, Q5, Q7, Q8, Q9, and Q10.

# payment-processor 500 errors · 0.95 → AUTO-EXECUTE

```
$ python query_live.py "payment processor returning 500 errors, transactions not completing at checkout"

┌──────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │    0.495 ✗ weak match │ [INC0043135] payment-processor 5xx spike (70%)  │
│ 2 │ incidents.json │    0.492 ✗ weak match │ [INC0043231] payment-processor 5xx spike (72%)  │
│ 3 │ incidents.json │    0.488 ✗ weak match │ [INC0043183] payment-processor 5xx spike (71%)  │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.495. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : rollback-to-previous-version
  Confidence       : 0.95
  Decision Gate    : AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : Three prior payment-processor 5xx spikes all resolved by rollback.
                     Standard mitigation confirmed despite weak RAG similarity scores.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  rollback-to-previous-version  (conf 0.95 ≥ 0.70 gate)  │
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 8.2s
```

LIVE OUTPUT — QUERY 4 / 10

# inventory sync stalled · 0.66 → PENDING REVIEW

```
$ python query_live.py "inventory sync stalled, stock levels on website not updating after warehouse batch job"

┌──────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.675 ~ partial match │ [INC0041047] inventory-sync batch job failed     │
│ 2 │ incidents.json │  0.656 ~ partial match │ [INC0043477] inventory-sync pipeline stall 75min │
│ 3 │ incidents.json │  0.639 ~ partial match │ [INC0042043] inventory-sync failing 14 stores    │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

✓ Strong precedent found — incidents.json (similarity 0.675)

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : force-inventory-sync
  Confidence       : 0.66
  Decision Gate    : PENDING REVIEW
  AI Source        : ollama:qwen3:14b
  Reasoning        : Stalled sync mirrors INC0043477 pipeline-stall scenario exactly.
                     force-inventory-sync would resolve — confidence just below gate threshold.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ■  PENDING HUMAN REVIEW  —  force-inventory-sync  (confidence 0.66 < 0.70 gate)     |
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 15.4s
```

LIVE OUTPUT — QUERY 5 / 10

# store POS tills unresponsive · 0.75 → AUTO-EXECUTE

```
$ python query_live.py "store POS system tills unresponsive, cashiers cannot process customer transactions"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.573 ~ partial match │ [INC0043165] store-pos-system pipeline stall  │
│ 2 │ incidents.json │  0.569 ~ partial match │ [INC0043333] store-pos-system pipeline stall  │
│ 3 │ incidents.json │    0.547 ✗ weak match  │ [INC0041062] store-pos-system API latency spike │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.573. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : rollback-to-previous-version
  Confidence       : 0.75
  Decision Gate    : ABOVE GATE → AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : P1 incident — Consul mesh misconfiguration caused 2,300-store outage.
                     Configuration rollback was the confirmed resolution. Pattern matches.

┌──────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  rollback-to-previous-version  (conf 0.75 ≥ 0.70 gate)  │
└──────────────────────────────────────────────────────────────────────────┘
Total runtime: 9.9s
```

LIVE OUTPUT — QUERY 7 / 10

# order fulfilment latency spike · 0.75 → AUTO-EXECUTE

```
$ python query_live.py "order fulfilment delays caused by upstream pricing engine latency spike during peak hours"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.571 ~ partial match │ [INC0043496] order-fulfilment P99 breach (1102ms)│
│ 2 │ incidents.json │  0.565 ~ partial match │ [INC0043448] order-fulfilment P99 breach (4926ms)│
│ 3 │ incidents.json │  0.558 ~ partial match │ [INC0043304] order-fulfilment P99 breach (798ms) │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.571. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : scale-up-replicas
  Confidence       : 0.75
  Decision Gate    : ABOVE GATE → AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : Upstream latency causes downstream throughput pressure.
                     Scaling replicas reduces per-instance load while root cause is addressed.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  scale-up-replicas  (confidence 0.75 ≥ 0.70 gate)     │
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 12.2s
```

LIVE OUTPUT — QUERY 8 / 10

# notification service duplicate emails · 0.35 → PENDING REVIEW

```
$ python query_live.py "notification service sending duplicate confirmation emails to customers after every order event"

┌──────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬──────────────────────┬──────────────────────┬──────────────────────────────────────┐
│ # │ Source               │        Similarity    │ Excerpt                              │
├───┼──────────────────────┼──────────────────────┼──────────────────────────────────────┤
│ 1 │ incidents.json       │  0.687 ~ partial match │ [INC0041294] notification-svc duplicate  │
│   │                      │                      │ PUSH notifications (order-fulfilment retry)│
│ 2 │ incidents.json       │    0.520 ✗ weak match  │ [INC0041158] notification-svc email queue  │
│ 3 │ INC0042468-notif-...md │   0.467 ✗ weak match  │ Auto-generated runbook: unsubscribe links  │
└───┴──────────────────────┴──────────────────────┴──────────────────────────────────────┘

✓ Strong precedent found — incidents.json (similarity 0.687)
  [note: matched push notifications, not email — semantic mismatch]

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : restart-service-with-memory-bump
  Confidence       : 0.35  ← MODEL SELF-DOWNGRADED
  Decision Gate    : RULE-BASED FALLBACK
  AI Source        : ollama:qwen3:14b
  Reasoning        : Runbook context (INC0042468) used restart-with-memory-bump for config issue.
                     Current incident involves email duplicates, not push notifications.
                     Semantic mismatch between incident type and available context.

┌──────────────────────────────────────────────────────────────────────────────────┐
│ ▪  PENDING HUMAN REVIEW  —  restart-service-with-memory-bump  (conf 0.35 < 0.70)    │
└──────────────────────────────────────────────────────────────────────────────────┘
Total runtime: 22.5s
```

# supplier EDI orders dropped · 0.65 → PENDING REVIEW

```
$ python query_live.py "supplier integration service dropping EDI purchase orders during XML schema validation"

┌──────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.705 ~ partial match │ [INC0041737] supplier-integration EDI batch  │
│   │                │                      │ failure — Supplier-31 schema version mismatch │
│ 2 │ incidents.json │  0.561 ~ partial match │ [INC0043383] supplier-integration ETL failure │
│ 3 │ incidents.json │    0.542 ✗ weak match  │ [INC0041195] supplier-integration EDI timeout │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

✓ Strong precedent found — incidents.json (similarity 0.705)

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : force-inventory-sync
  Confidence       : 0.65
  Decision Gate    : PENDING REVIEW
  AI Source        : ollama:qwen3:14b
  Reasoning        : EDI orders dropped during XML schema validation matches INC0041737 closely.
                     force-inventory-sync addresses data pipeline stalls, not schema mismatches.
                     Plausible mitigation, but not the correct root-cause fix. Correctly held.

┌──────────────────────────────────────────────────────────────────────────┐
│ ▪ PENDING HUMAN REVIEW  —  force-inventory-sync  (confidence 0.65 < 0.70 gate)  │
└──────────────────────────────────────────────────────────────────────────┘
Total runtime: 25.7s
```

# recommendation engine degraded · 0.45 → PENDING REVIEW

```
$ python query_live.py "recommendation engine returning identical product suggestions for all users after overnight model retraining"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │    0.393 ✗ weak match │ [INC0041839] product-search autocomplete empty   │
│ 2 │ incidents.json │    0.356 ✗ weak match │ [INC0043065] product-search feature flag misconf │
│ 3 │ incidents.json │    0.315 ✗ weak match │ [INC0043017] product-search feature flag misconf │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.393. AI will deduce a remediation.
  [all three matches are from a different service; zero relevant knowledge base entries]

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : reindex-elasticsearch
  Confidence       : 0.45  ← ACKNOWLEDGING LIMITS
  Decision Gate    : RULE-BASED FALLBACK
  AI Source        : ollama:qwen3:14b
  Reasoning        : Identical suggestions indicate stale/improperly updated index data.
                     No recommendation-engine incidents in knowledge base. Cross-service inference
                     only. Model self-downgraded — this is the system acknowledging its limits.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ■  PENDING HUMAN REVIEW  —  reindex-elasticsearch  (confidence 0.45 < 0.70 gate)   │
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 13.8s
```